# ODT Summative 2

## Idea Brief :

**Smart Locker System** :

A multi-gateway security locker, encompassing physical hardware and a mobile application. The first gateway uses an OTP generated on the app, which you enter on the safe to unlock the door. The second gateway challenges you with a math equation, and its answer is the exact distance at which you place your hand for the sensor to detect it and open the gate. The third gateway uses a coloured pattern inside the safe to open another mini locker.

| PHYSICAL STRUCTURE | DESCRIPTION |
|---|---|
| **Main Locker** | Primary enclosure with the core locking mechanism, push buttons, NeoPixel, buzzer, servo motor, door sensor, and ultrasonic sensor. |
| **Mini Locker** | Smaller compartment physically attached beside the main locker, with independent locking mechanism (dc motor), only accessible after the main locker is opened. |
| **Back Wall (main locker)** | The interior back wall of the main locker has 4 coloured sheets (physical, non-electronic) fixed to it, revealed only once the main door swings open, serving as the visual clue for the mini locker sequence. |
| **Distance Scale** | A physical scale placed in front of the locker, so that users can position their hand at the correct distance during Gateway 2. |

## Gateway 1 –OTP

### CONCEPT

A **one-time password** is generated each session from the MIT App Inventor application. The user

reads the OTP from the app and enters it physically on the locker using push buttons.

### How It Works

- The app generates a randomised numeric OTP (within a defined range/digit count) each time the user requests one.
- The OTP is transmitted to the ESP32 over Bluetooth.
- The user enters the code digit-by-digit using push buttons labelled 1-9 on the locker face.
- The ESP32 compares the entered sequence against the received OTP in real time.

### Push button Sound Feedback

Each button press triggers a short, distinct frequency tone from the buzzer, providing the user with audio confirmation of each digit entered.

| **COMPONENT** | **CORRECT ENTRY RESPONSE** | **WRONG ENTRY RESPONSE** |
|---|---|---|
| NeoPixel | Lights up green (sequentially) | Lights up red (sequentially) |
| Servo Motor | Rotates to open position [PWM] | No response |
| Buzzer | **Door sensor** detects door physically swinging open, triggers buzzer at frequency 1 | No door sensor response |

### Lockout Condition

After 3 consecutive wrong attempts, the entire Gateway 1 input system locks, the push buttons stop responding and no further entries are accepted.

## Gateway 2 – Math Equation w/ Ultrasound sensor distance

### CONCEPT

Instead of entering a code, the user must solve a math problem and position their hand at that distance from an ultrasonic sensor mounted on the locker.

### How It Works

- The app generates a simple arithmetic equation whose answer always falls within a defined range, ideally **5 to 20**(e.g. "3 × 4 + 2 = ?", answer 14).
- The equation is displayed on the app interface.

- A physical scale or ruler is affixed to the locker beside the ultrasonic sensor, marked in centimetres.
- The user solves the equation mentally, then places their hand at the corresponding centimetre mark in front of the sensor.
- The ESP32 reads the ultrasonic distance.
- If correct → same success response as Gateway 1 (green NeoPixel, servo opens, door sensor triggers buzzer)
- If wrong → same rejection response (red NeoPixel, rejection buzzer, no servo)

## Gateway 3 - Colour Sequence [Mini Locker]

### CONCEPT

Can only be unlocked after the main locker has already been opened, using a colour sequence revealed on the interior back wall.

### How It Works

- Once the main locker door opens (via Gateway 1 or 2), it reveals the interior back wall covered with 4 coloured sheets
- This sequence of colours is the "key" to the mini locker
- The MIT App interface contains buttons in the same colours as the wall, whose purpose, prior to the opening of the locker, was deliberately kept vague
- The user presses the coloured buttons in the same sequence as the colours on the wall
- Correct → mini locker servo opens
- Wrong → rejection response on NeoPixel and buzzer

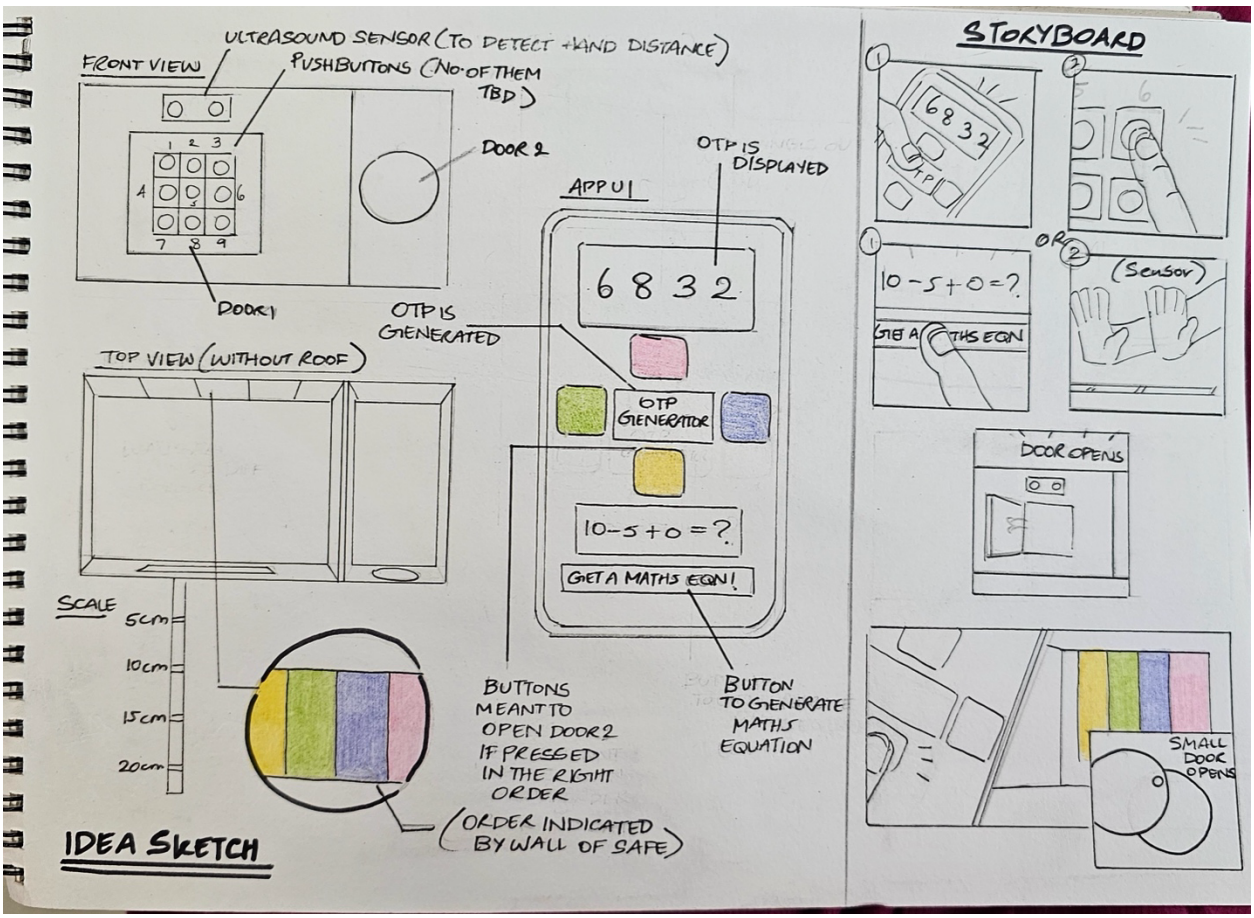
## App Interface

| COMPONENT | FUNCTION |
|---|---|
| OTP generation button | Random four digit OTP is generated when button is pressed to be entered into the safe |
| Maths equation generator | Random maths equation is generated, whose answer is the distance in cm at which the user's hand is to be kept for the Ultrasound sensor to sense it |
| Coloured buttons (4) | When pressed in the right order will unlock the mini locker |
| OTP Display | Displays the OTP generated on pressing the button |
| Equation Display | Displays the equation generated on pressing the button |

### How it works

When OTP button is pressed, the code **generates a random OTP** and sends it to the phone **via Bluetooth**, where it appears on the display. Entering this OTP using push buttons opens the door.

When pressing the equation generator button, an equation is generated and displayed. When this **equation is solved and your hand is placed at that distance**, the sensor senses it and the door of the main locker opens.

When the coloured buttons are pressed **according to the pattern on the wall inside the safe**, the mini locker door opens.

## Basic Idea Sketch :